

Standard Operating Procedure

Utility Strike Prevention – In-Floor Heating, Concealed Services & Wall-Mounted Equipment

Document No.	SOP-LV-006
Related Documents	SOP-LV-001, SOP-LV-002, SOP-LV-003, IR-2026-07-08-001
Effective Date	July 13, 2026
Revision	Rev. 0
Jurisdiction	Ontario (O. Reg. 213/91 – Construction Projects; WSIB)
Applies To	All field technicians and subcontractors performing drilling, anchoring, or penetration work into floors, walls, or ceilings
Approved By	_____

1. Purpose

This procedure establishes mandatory steps to identify and avoid concealed in-floor heating, plumbing, electrical, and other building services before drilling, coring, anchoring, or otherwise penetrating floors, walls, or ceilings. It was issued following a July 8, 2026 incident at Elsie McGill Public School in which a technician struck an in-floor heating pipe while drilling (Incident Report IR-2026-07-08-001).

2. Scope

This SOP applies to all low-voltage and structured cabling work performed by the company across Ontario (and Manitoba where applicable), including but not limited to: concrete floor drilling and coring, wall anchoring for TV mounts, brackets, and equipment enclosures, and any penetration into ceilings, walls, or slabs at commercial, institutional, or school sites.

3. Responsibilities

Site Supervisor

- Confirms customer/facility contact has been asked about concealed services before work is scheduled or begins.
- Obtains and reviews available as-built, mechanical, or electrical drawings for the work area.
- Authorizes use of non-destructive detection equipment where required.
- Has authority to stop work if concealed services cannot be confirmed.

Field Technician

- Does not drill, core, or anchor into any floor, wall, or ceiling until concealed-service verification for that specific location is complete.
- Reports any uncertainty, unexpected resistance, or suspected utility contact immediately and stops work.
- Completes and signs off on this procedure prior to starting affected work.

4. Customer Notification & Utility Verification (All Drilling Work)

Before any drilling or coring into a concrete floor, slab, or wall, the technician or supervisor must ask the customer or facility representative — in person, by phone, or in writing — whether any of the following are present in or near the work area:

- In-floor or in-slab radiant heating (hydronic or electric)
- Plumbing lines, drains, or supply piping embedded in the floor or wall
- Electrical conduit, wiring, or panels within the floor, wall, or ceiling cavity
- Post-tension cables or embedded structural reinforcement
- Fire suppression or sprinkler piping

This confirmation must be documented (email, work order note, or signed checklist) before drilling begins. Verbal-only confirmation with no record is not acceptable for school, hospital, or other institutional sites.

Where the customer cannot confirm the presence or absence of concealed services, or where drawings are unavailable, the work area must be treated as unverified and Section 5 or Section 6 precautions apply before any penetration is made.

5. In-Floor Heating Precautions

STOP WORK CONDITION:

Do not drill into any floor area where in-floor heating is present, suspected, or unconfirmed.

Before drilling into any concrete or slab floor:

- Ask the customer directly whether the floor has in-floor/radiant heating, and get this in writing where possible.
- Review mechanical drawings for the space, if available, to identify heating loop layout and manifold locations.
- Where in-floor heating is confirmed present, obtain the heating loop layout before selecting drill locations, and maintain clearance from marked loop zones per the layout.
- Where heating cannot be ruled out (older buildings, missing drawings, renovated spaces), use a non-destructive method to verify before drilling: thermal imaging camera (with heating system running), utility/rebar locator, or a small pilot hole with immediate withdrawal if resistance or moisture is encountered.
- If a pipe, moisture, or unexpected material is encountered while drilling, stop immediately, withdraw the bit, and do not continue at that location.

Reference: this section directly addresses the failure mode identified in IR-2026-07-08-001, where drilling proceeded without confirming in-floor heating layout.

6. Wall-Mounted Equipment & TV Mount Precautions

STOP WORK CONDITION:

Do not anchor into a wall without confirming what is on the opposite side.

Before drilling or anchoring for a TV mount, bracket, or wall-mounted enclosure:

- Identify what room or space is on the other side of the wall. Treat any wall shared with a washroom, kitchen, kitchenette, janitorial closet, or mechanical room as a probable wet wall.
- Scan the wall with a stud finder capable of detecting metal and AC wiring, not just studs, across the full mounting area before drilling any hole.
- Where a wet wall or electrical run is suspected, avoid drilling within the typical plumbing stack or wiring zone and, where possible, confirm exact pipe/conduit routing with the customer or a facilities contact before proceeding.
- For masonry, concrete block, or slab walls, apply the same verification steps as Section 4 and 5 before anchoring.
- If in doubt, select an alternate mounting location or install a backer board fastened only to confirmed stud locations rather than drilling into an unverified wall cavity.

7. Stop-Work Authority

Any technician has the authority and the obligation to stop drilling or anchoring work at a location if concealed services cannot be confirmed, if documentation is incomplete, or if signs of a possible utility (moisture, resistance, exposed conduit, unusual odour) are encountered. Work does not resume at that location until the condition is resolved by the Site Supervisor.

8. If a Strike Occurs

- Stop work immediately at the location.
- Make the area safe (de-energize if electrical, shut off supply if plumbing, ventilate if applicable) without attempting repairs.
- Notify the Site Supervisor immediately.
- Do not disturb the site beyond what is needed for safety, pending assessment.
- Complete an incident report per company procedure and notify the facility/customer contact.

9. References

- Ontario Regulation 213/91 – Construction Projects (Occupational Health and Safety Act)
- WSIB – Workplace Safety and Insurance Board reporting requirements
- Canadian Electrical Code (CEC) / CSA standards, as applicable

- National Building Code (NBC) of Canada, as applicable
- SOP-LV-002 – Concrete Floor Drilling with Utility Strike Prevention
- SOP-LV-003 – Ontario Jobsite Health & Safety Checklist
- IR-2026-07-08-001 – Incident Report, Elsie McGill Public School (July 8, 2026)

10. Acknowledgement & Sign-Off

By signing below, the technician confirms they have read, understood, and agree to follow this procedure on every applicable job, including asking the customer about concealed services before drilling and verifying the opposite side of any wall before mounting equipment.

Role	Printed Name / Signature	Date